Integration of Phenolic Secondary Metabolism into a Grapevine Genome-Scale Metabolic Model Using DeePlants

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Abstract. .

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1 Motivation

In grapevines, secondary metabolites are generally not considered essential for basic plant growth, but are crucial for its defence and, from an agricultural point of view, important for wine quality [6]. These compounds are primarily affected by the environment, such as light and temperature and by viticultural practices [8]. Genome-scale metabolic models (GSMM) are important tools used to represent plant's metabolic network. The study will involve the *Vitis vinifera* GSMM iMS7199, featuring secondary metabolites such as anthocyanins, quercetin, resveratrol, terpenoids, carotenoids, cholesterol, and diacylsucrose [13]. Through the integration of the model, we will be able to predict how the plant redistributes its resources to produce phenolic compounds, the main group of plant secondary metabolites that highly contribute to wine flavor. To predict how the plant redistributes its resources to produce phenolic compounds, the main group of plant secondary metabolites that highly contribute to wine flavor, the model will be integrated with a computational framework, DeePlants, aiming to improve the elucidation of secondary metabolism, as well as the identification and functional mapping of secondary metabolites and biosynthetic genes. This tool will allow the expansion of metabolic pathways that are poorly represented in databases. To this end, the aim of the present work is to integrate metabolite presence and absence data on phenolic compounds from Syrah grapes into a grapevine GEM, and to use DeePlants to annotate specific reactions and metabolites of anthocyanins, flavan-3-ols, flavonols, and phenolic acids. Furthermore, it aims to simulate the distribution of metabolic fluxes under different environmental and ripening scenarios, thereby validating the model's predictions against the observed metabolite profiles to evaluate both the model's accuracy and biological insight.

2 State of the art

2.1 Plants secondary metabolites

Plants yield a vast array of secondary metabolites that play a fundamental role in mediating environmental interactions and stress adaptation [17], with over 200,000 different known compounds [16]. Classified into three main groups, terpenoids, alkaloids, and phenolic compounds (PCs) [3], they play crucial roles in plants, such as regulation of growth and development, and respond to diverse biotic and abiotic stresses, such as pathogens and herbivores [17].Unlike primary metabolites, their production is typically triggered by specific environmental stimuli or during specific developmental stages, to increases their chances of survival [17]. The current literature demonstrates that plants use these compounds to manipulate their surrounding environment through plant-plant, plant-animal, and plant-microorganism interactions [16]. Despite exhibiting diverse biochemical properties, the best characterized and most widely highlighted is their antioxidant activity [5]. If antioxidant enzymes cannot sufficiently neutralize oxidation, secondary metabolites such as PCs become critical for limiting reactive oxygen species (ROS) accumulation [17].

2.2 Phenolic compounds

PCs are generally present in most plant tissues, such as fruits, seeds, leaves, stems, roots, among others[12]. They are composed of at least one aromatic ring with a hydroxyl group [12] and present great structural diversity, exceeding 8,000 PCs per plant [12]. They can be divided into three main types: polyphenols, including lignin and tannins, oligophenols, such as flavonoids, stilbenes, and coumarins, and phenolic acids, including derivatives of benzoic acid and cinnamic acids [10]. PCs represent the second largest group of organic compounds in the plant kingdom, playing fundamental roles in plants, such as providing structural support and protection against UV radiation, biotic and abiotic stress. They also play a crucial role from a consumer perspective, through their antioxidant activity, as well as through their involvement in the regulation of various cellular processes. Furthermore, they influence the quality attributes of fruits and vegetables, including bitterness, colour, and flavour [12].

In plants, PCs biosynthesis proceeds via shikimic acid pathway [19], where carbohydrates from glycolysis and the pentose phosphate pathway are converted into chorismate, a precursor of tryptophan and phenylalanine. In the case of phenylalanine, it initiates the phenylpropanoid pathway through a deamination reaction catalyzed by phenylalanine ammonia-lyase (PAL), yielding trans-cinnamic acid. This enzymatic reaction represents a crucial regulatory step in the production of numerous phenolic compounds. Intermediate metabolites in this pathway are also converted into phenolic acids [10].

Over the last decade, at least 138 PCs have been identified in the vegetative organs of the *Vitis vinifera*, including stilbenes, anthocyanins, hydroxycinnamic acids, hydroxybenzoic acids, flavanones, flavones, flavonols, flavan-3-ols, and coumarins. These compounds are distributed throughout the grapevine, with flavonols and flavan-3-ols being the most represented in the leaves and stems [7]. In grape berries, PCs are primarily present in the skins and seeds, and flavonols are the most abundant PCs in grape skins [9].

PCs contribute to the perception of wine quality, influencing the color, for browning reactions as well as for various reactions with anthocyanins that lead to color stabilization in red wines [9][18], flavor, influencing sensory attributes like bitterness and astringency [4], and sensory characteristics of the wine. Polyphenols, particularly certain phenolic acids and flavonols, participate in the copigmentation phenomenon, which is why anthocyanins exhibit a much more intense color than would be possible without them [9][18].

2.3 *Vitis vinifera* model

Through the integration of tissue-specific omics data for the grapevine leaf, stem, and grapes, a diel multi-tissue model of *V. vinifera* was constructed by Sampaio and collaborators [13].

The study involves the genome-scale metabolic model (GSMM) iMS7199, specifically reconstructed for the grapevine *V. vinifera*. The genome version PN40024.v4 was used for this study. This model presents 5399 reactions with 1624 transporters and 244 exchanges, and 5141 metabolites, distributed across eight compartments, these being cytosol, mitochondria, Golgi apparatus, endoplasmic reticulum, chloroplast, peroxisome, vacuole, and extracellular space[13]. Since genomic annotation was performed using protein sequences, the Gene-Protein-Reaction (GPR) rules in this model were established using protein identifiers rather than gene identifiers. The model includes 7199 protein identifiers that represent the 6018 genes of *V. vinifera* genome [13].

The secondary metabolite biosynthesis, derived from the repository containing metabolic information from the PlantCyc 14.0 and MetaCyc 26.1 databases, and the Universal Protein Resource (UniProt), is the metabolic pathway class associated with the highest number of unique reactions, around 140. Compared to previous plant models *V. vinifera*'s GSMM represents a major advancement, especially for secondary metabolism, which is frequently underrepresented in plant models [13].

Through this model and using the integration of RNA-Seq data, specific models were created for different parts of the plant, such as the stem, the leaf, and the grape berry, with the latter being further divided into green and mature developmental stages. To make the simulation more realistic, the models were merged into a diel multi-tissue model, which simulates the light and dark phases. The model allowed the simulation and investigation of the grapevine's metabolic behavior under different phototrophic and heterotrophic conditions. With the aim of identifying the associated metabolic pathways and gene annotation, these reactions were analyzed and secondary metabolite biosynthesis pathways were the ones that obtained the highest number of unique reactions. The main limitation is the lack of experimental data to define pseudo-reactions representing the consumption of macromolecules for biomass synthesis. The model can be integrated with other datasets to investigate condition-specific metabolic phenotypes, providing a better understanding of plant metabolism [13].

FBA and pFBA Methods such as Flux Balance Analysis (FBA) and parsimonious FBA (pFBA) are required to verify the ability of the GSMM iMS7199 to accurately simulate key metabolic processes, including photosynthesis, photorespiration, and respiration[13].

FBA is a metabolic modelling approach that uses a set of physical, biochemical, and thermodynamic constraints to predict steady-state metabolic fluxes from stoichiometric models [14], where stoichiometric models are the mathematical representations of the organism's metabolic reactions, in which the model assumes a steady state, where the rate of change of metabolite concentrations over time is zero, and metabolic fluxes describe the rates at which each metabolite is consumed or produced by each reaction in the network[11]. Since it is based on linear equations, FBA is computationally efficient and widely used to simulate flux distributions in extensive metabolic networks. This is commonly used in the study of plant metabolism [14], and through these calculations, it is possible to determine an optimal flux distribution for a given objective function [13]. This has been widely used to simulate GSMMs. However, in the solution space for a particular objective, there are generally multiple optimal solutions. In this model, pFBA is employed to simulate plant secondary metabolism, an extension of FBA which, by selecting a flux distribution from the optimal FBA solution space, minimizes the total sum of fluxes [13].

2.4 DeePlants

Retrobiosynthesis The biosynthetic pathways of more than 90% of natural products (NPs) are still undiscovered. Despite their complex and highly diverse structures, NPs are synthesised from a few simple biological building blocks. For this porpose, retrobiosynthesis is employed, which starts from a specific target molecule and aims to identify a viable and efficient synthetic pathway, going through a large number of possible chemical reactions until ultimately reaching the precursor molecules [20]. Given the performance of reinforcement learning, the use of Monte Carlo Tree Search (MCTS) guided by neural networks was proposed for retrosynthetic planning. This approach is fundamental, as it enables the creation of a self-improving learning process for the neural networks guiding the search algorithms, allowing them to assess a given state and predict the most efficient and viable reactions to attempt next [15][20].

Identifying these synthetic routes is crucial and computer-assisted retrosynthesis is essential to solve it. The MEEA* algorithm offers a powerful solution by combining the heristic algorithms MCTS and A*, allowing it to escape local optimal branches and preventing exhaustive exploration of all branches [20].

MCTS operates through a balance between exploitation, where it takes advantage of and expands upon paths that have already been tested and which, according to current scores, appear to be the most promising and efficient, and exploration, where new and unknown paths are tested in the research tree, ensuring that potentially better alternative routes are not overlooked. The algorithm operates through a search process that iterates through four steps. The first is selection, where the algorithm traverses the search tree, used by algorithms to map the various possible chemical synthesis pathways of a molecule, through two parts, the nodes and the edges. The nodes represent the states of the synthesis. The initial node, the root of the tree is the target molecule and the nodes below it contain the set of intermediate molecules and building blocks present at that point in the synthesis. The edges are the connections between the nodes. Each edge represents a chemical reaction that causes the transition from the parent node, the products, to the child node, the reactants. This step uses an exploratory tree policy like pUCT, which calculates and ensures a balance between exploring new options and exploiting those that already appear promising. The second step is the expansion, where new promising nodes, suggested by a retrosynthetic model that predict the most promising chemical reactions for breaking down a target molecule into its possible precursors, are incorporated into the tree. The third step is evaluation, estimating the subsequent effort required to reach the goal from the new node via rapid simulations, rollouts, or neural network-based predictive models. Finally, the update step, known as backpropagation, adjusts the values of all nodes along the traversed path based on the leaf node's evaluation, improving subsequent node selection [20].

The A* algorithm is a heuristic search method that finds an optimal solution when the heuristic function is admissible (and consistent). It operates by iteratively evaluating nodes using the function $f(s) = g(s) + h(s)$, where $g(s)$ is the cost from the start node to state s, and $h(s)$ is an estimate of the cost from s to the goal. The algorithm maintains two sets: *OPEN*, containing nodes to be explored, and *CLOSED*, containing already explored nodes. At each step, A* selects the node from *OPEN* with the lowest f-value. If this node is the goal, the search terminates successfully. Otherwise, the node is expanded, its successors are generated, and their costs are updated if a better path is found before adding them to *OPEN* [20]. This algorithm plays a crucial role, serving as a search engine to identify bio-retrosynthetic planning pathways [20].

The MEEA* algorithm adds a Candidate Set to the *OPEN* and *CLOSED* lists, operating in three vital phases. The simulation phase uses a purely MCTS exploratory policy for look-ahead search without permanent tree expansion. The reached nodes are evaluated, placed in the Candidate Set, and path values are updated. In the selection phase, MEEA* applies A* to the Candidate Set, choosing the state with the lowest $f(s)$ value. Finally, during expansion, the selected node is submitted to the retrosynthetic prediction model, and its children are permanently integrated into the search tree. Tests on the USPTO benchmark revealed that MEEA* achieved strong performance. When replacing Retro* with MEEA*, the success rate for identifying retrosynthetic pathways for NPs compounds increased from 88.66% to 97.68%. Even with a massive advantage, Retro* stagnated at 93.04%, remaining below MEEA*'s standard results [20].

Genome annotation methods Accurate and efficient prediction of enzyme function is essential for genomic annotation workflows. These predictions are essential, and considering the small fraction of enzymes with assigned EC (Enzyme Commission) numbers, numerous tools that use similarity and Machine Learning (ML) techniques have been developed. BLASTp (Basic Local Alignment Search Tool for protein sequences) is the most widely used gold standard tool, but it offers no way of assigning a function to proteins without known homologous sequences. With the advent of Deep Learning (DL), these approaches became better suited for tasks involving large volumes of data. To overcome BLASTp's limitations, DL models began using protein Large Language Models (LLMs) [2].

To evaluate the best approach for genome annotation, Capela et al. (2025) assessed the performance of the ESM2, ESM1b and ProtBERT language models in predicting EC numbers compared to BLASTp, to each other, and to models that rely on one-hot encodings of amino acid sequences [2]. In general, BLASTp outperforms DNN-LLM models on large datasets with homologous sequences available in databases, however, it should be noted that its advantage stems from being the most widely adopted annotation approaches. Despite this, DL models trained with LLMs achieved excellent results, with a very low difference in precision and recall tests compared to BLASTp. In scenarios where the training set had a scarcity of homologous sequences, LLMs outperformed BLASTp. DL models should be used when BLASTp fails to find a sequence with more than 25% identity in the training set, as its performance declines and it is then outperformed by DL models. Combining BLASTp with DNN-LLM models surpasses the performance of either tool used individually, providing an advantage in mainstream annotation routines and more complex tasks [2].

Precursor prediction For precursor prediction, the use of the precursor tool will be essential. The approach to be taken involves an automated ML and DL framework used to optimize molecular analysis tasks. Through its AutoML engine, DeepMol automates the search for the optimal pipeline. It is utilized to solve a chemical and biological problem by automatically identifying the best computational model to predict the precursors of plant secondary metabolites [1].To ensure the identification of the best model, it is described that DeepMol was used to test and compare various approaches, and for this, ML models need to process the data by providing them with a standard representation called molecular fingerprints (FPs) and the main approaches to generate them are Morgan FPs and Layered FPs. Among the approaches designed for NPs, there is the a type of FPs that uses substrings known as SMILES, a pre-existing format for writing the three-dimensional structure of a molecule in a simple line of text[1]. Capela et al. (2025) demonstrated that the models identified by DeepMol's AutoML present an advantage over models that use NP-focused FPs. During the testing of the model, precursors were predicted, achieving success for over three-quarters of the compounds. This is an approach that will accelerate the exploration of biosynthetic pathways, potentially expediting the discovery and understanding of natural product biosynthesis [1].

3 Next steps

- Perform compound retrobiosynthesis using DeePlants to identify precursors and their associated metabolic pathways.
- Perform genome annotation by predicting EC numbers using BLASTp and DNN-LLM.
- integrate predictions of metabolic pathways into the *Vitis vinifera* iMS7199 genome-scale metabolic model (GSMM).

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